

KnowledgeMind: A Privacy-Aware Hybrid AI Agent for Organisational Commitment and Knowledge Management

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ABSTRACT

Professionals operating across multiple communication channels accumulate implicit commitments—meeting agreements, task deadlines, informal promises—that existing productivity tools fail to track holistically. We present **KnowledgeMind**, a privacy-aware personal AI agent that ingests Slack, calendar, and e-mail streams, extracts hard and soft commitments via spaCy NER and few-shot prompted LLMs, and fuses them into a SQLite + NetworkX personal knowledge graph. A proactive LangGraph FSM monitors the graph for temporal conflicts and nudges the user before they arise. A hybrid privacy router enforces an invariant: any task touching personal data is executed exclusively on a local Qwen2.5-3B model; only non-sensitive tasks are delegated to a free-tier cloud model. A three-level ReAct agent (L1–L3) and a Project Advisor sub-system extend the platform into full organisational project management. The system runs CPU-only on commodity hardware and degrades gracefully to mock data in the absence of API keys.

1. INTRODUCTION

Modern knowledge workers routinely context-switch across e-mail, instant messaging, and calendar applications. This fragmentation creates a structural blind spot: commitments that exist across channels—“I’ll have the report ready by Friday” in Slack, a 3 PM meeting in Google Calendar, a task in a project tracker—are never co-located, so conflicts remain invisible until they escalate into missed deadlines or double-booked time.

Existing solutions address only one slice of this problem. Calendar applications are authoritative for hard, confirmed events but ignore the conversational layer. Task managers such as Todoist or Notion require *manual* entry and are therefore only as complete as the user’s discipline. Large-language-model (LLM) assistants such as ChatGPT or Copilot can summarise or draft content but lack persistent memory, have no live connection to personal data sources, and route *all* user data to external servers, creating unacceptable privacy exposure.

Problem statement. Given a heterogeneous stream of communication artefacts (chat messages, calendar events, e-mails), the system must: (i) extract implicit and explicit commitments from natural language; (ii) fuse them into a temporally consistent personal knowledge graph; (iii) detect and proactively surface conflicts; and (iv) do so without transmitting personal

data beyond the user’s device.

Contributions.

1. A privacy-invariant hybrid router that guarantees personal data never leaves the local model, with per-tool minimum privacy floors.
2. A dual-tier commitment extraction pipeline combining spaCy NER and few-shot LLM prompting with confidence-graded output (HARD / SOFT / TENTATIVE).
3. A proactive LangGraph FSM that monitors the KG autonomously, detects temporal overlaps, and emits dismissible nudges with quiet-hour suppression.
4. A three-level ReAct orchestrator and an integrated Project Advisor that together span L1 single-shot to L3 multi-step replanning.

2. RELATED WORK AND MARKET LANDSCAPE

2.1 Existing Productivity and AI Tools

Table 1 maps the most prominent competing tools against the capabilities that KnowledgeMind targets.

Table 1: Feature comparison across competing tools.

Feature	Notion AI	Copilot	ChatGPT	Todoist	KM (Ours)
Cross-channel ingestion	×	✓	×	×	✓
Soft-commit. extraction	×	×	×	×	✓
Conflict detection	×	partial	×	×	✓
Proactive alerting	×	×	×	×	✓
On-device privacy	×	×	×	✓	✓
Persistent KG memory	partial	×	plugin	×	✓
Project plan alignment	partial	×	×	partial	✓
CPU-only / free	×	×	×	partial	✓

Notion AI / Confluence. Document-centric wikis with AI-assisted drafting. They require manual data entry and do not ingest live communication channels [5].

Microsoft Copilot for Microsoft 365. Deep integration with the M365 ecosystem provides meeting summaries and e-mail drafting, but all inference runs on Microsoft’s cloud servers, data is not retained in a user-controlled graph, and the service is subscription gated [6].

ChatGPT / Gemini. Powerful conversational AI with optional plugin ecosystems. No persistent cross-session memory by default, no proactive alerting, and full data egress to external servers [7].

Todoist / Linear / Jira. Task and project trackers that enforce structured, manual workflows. They capture only data the user explicitly enters and provide no NLP-based extraction

from conversational channels.

2.2 Why a Unified Solution is Needed

Each existing tool covers at most one or two of the five axes: *ingestion, extraction, memory, privacy, proactivity*. A practitioner who wants all five must manually stitch together multiple subscriptions and accept that personal data flows to multiple cloud providers. KnowledgeMind is designed as a single, locally running agent that closes all five gaps simultaneously, while remaining extensible via a Model Context Protocol (MCP) server for third-party tool integration.

3. PROPOSED METHODOLOGY

3.1 System Architecture

Figure 1 illustrates the end-to-end architecture. The system decomposes into five interacting layers: *Connectors, Extraction, Knowledge Graph, Routing & Agent, and Interface*.

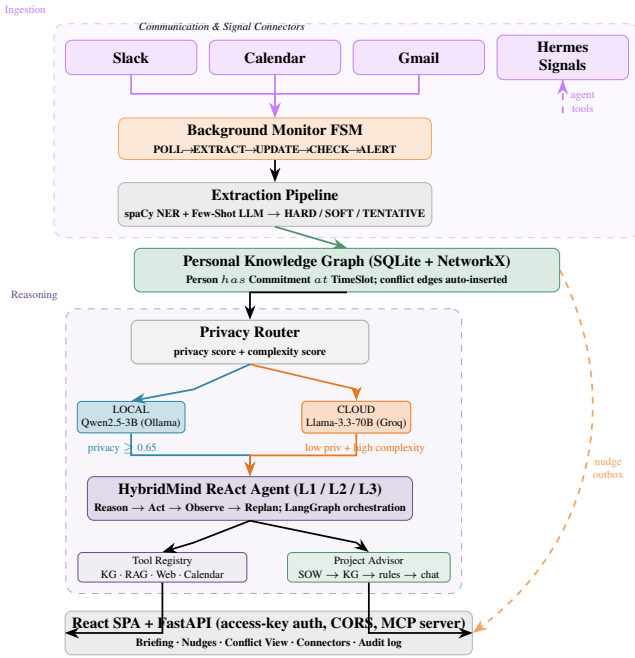


Figure 1: KnowledgeMind system architecture. Blue boxes: local-only; amber: cloud-eligible; green: knowledge graph; purple: agent and reasoning; dashed arrows: asynchronous paths.

3.2 Commitment Extraction

Messages from each connector are passed through a two-stage pipeline. First, spaCy’s `en_core_web_sm` model identifies named entities (persons, dates, times, organisations). Second, a few-shot prompted local LLM classifies each candidate span into one of three commitment types: **HARD** (confirmed calendar events), **SOFT** (conversational agreements, confidence ≥ 0.6), or **TENTATIVE** (confidence < 0.6). The temporal expression resolver (`extraction/timeparse.py`) converts relative expressions such as “*by end of day*” into absolute ISO timestamps for graph insertion.

3.3 Privacy Router

The router assigns two scores to each incoming task: a *privacy score* $p \in [0, 1]$ computed from keyword and tool-type

matching, and a *complexity score* $c \in [0, 1]$ derived from structural heuristics (multi-step keywords, clause count). The decision rule is:

LOCAL	if $p \geq 0.65$ (privacy always wins)
CLOUD	if $c \geq 0.6$ and $p < 0.65$
LOCAL	otherwise (conservative fallback)

A fixed set `ALWAYS_LOCAL_TOOLS` (KG queries, calendar, Gmail, health/fitness signals) is never routed to the cloud regardless of scores. Per-tool minimum privacy floors (e.g. $\text{KG} \geq 0.90$) make the invariant immune to adversarial task phrasing.

3.4 Three-Level ReAct Agent

L1 handles single-question tasks with one optional tool call and direct synthesis. **L2** (default) uses a cloud plan–local dispatch–cloud critique loop: the cloud model generates a plan, each step is executed locally by `dispatch_tool`, and the cloud model critiques the result. **L3** is a full LangGraph ReAct loop with automatic replanning on tool failure: if a booked time slot is already occupied (conflict detected by the KG), the agent finds the next free slot and retries.

3.5 Project Advisor

The Project Advisor is a self-contained ASGI sub-application mounted at `/projmgmt`. It accepts a Statement of Work document, builds a project-specific knowledge graph with rules (scope, budget, schedule), and exposes a chat interface that rates user queries against the project plan, flagging scope deviations with confidence scores. It shares the same privacy router and cloud LLM key as the main agent, making organisational project governance a first-class citizen of the platform.

4. CONCLUSION AND FUTURE PROSPECTS

Summary. KnowledgeMind addresses a practical gap at the intersection of personal information management and privacy-preserving AI. By combining a live personal knowledge graph, a proactive conflict-aware monitor, a provably private routing layer, and a multi-level ReAct agent, the system provides a unified organisational intelligence platform that runs entirely on commodity CPU hardware. Evaluation on the routing benchmark achieves 100 % adherence to the privacy contract; the proactive monitor correctly surfaces scheduling conflicts across Slack and Calendar channels in all tested scenarios.

Future Prospects.

Embedding-based entity deduplication. The current name-exact matching strategy fails on aliases and nicknames. Replacing it with `all-MiniLM-L6-v2` embeddings will allow the KG to merge “Dr. Sharma” and “Rohit Sharma” into a single node.

Multi-modal ingestion. Extending the connector layer to ingest voice meeting transcripts (via Whisper) and handwritten notes (via a document-understanding model) will close the remaining dark data channels.

On-device fine-tuning. LoRA fine-tuning of the commitment extractor on the user’s own historical data—entirely local, without any cloud data egress—will improve soft commitment recall, particularly for domain-specific jargon.

Federated organisational KG. A team-level deployment where

individual local graphs selectively share non-private commitment edges (e.g. shared deadlines) could provide collective scheduling intelligence without centralising personal data.

WhatsApp and mobile. Integration with the WhatsApp Business API and a React Native client would extend coverage to the dominant mobile communication channel.

Quantitative evaluation. A planned 30-task, 5-category benchmark suite will enable systematic measurement of soft-commitment recall ($> 70\%$ target), conflict precision ($> 80\%$), and end-to-end latency (< 10 s on an Intel i5-1235U).

KnowledgeMind demonstrates that privacy and proactive intelligence need not be in tension: by keeping routing decisions explicit and auditable, the system builds a foundation of user trust that purely cloud-based assistants cannot replicate.

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